

fuel and precursors for biosynthesis. For example, flowering plants stockpile the extra sugar in the form of starch, storing some in the chloroplasts and some in cells of roots, tubers, seeds, and fruits. In accounting for the use of food molecules produced by photosynthesis, note that most flowering plants lose leaves, roots, stems, fruits, and often their entire bodies to heterotrophs, including humans.

On a global scale, photosynthesis is the process responsible for the presence of O_2 in our atmosphere. Furthermore, although each chloroplast is minuscule, their collective productivity in terms of food production is enormous: Photosynthesis makes an estimated 150 billion metric tons of carbohydrate per year (a metric ton is 1,000 kg, about 1.1 tons). That's organic matter equivalent in mass to a stack of about 60 trillion biology textbooks—and the stack would

reach 17 times the distance from Earth to the sun! No chemical process is more important than photosynthesis to the welfare of life on Earth.

In Chapters 5 through 10, you have learned about many activities of cells. **Figure 10.22** integrates these cellular processes into the context of a working plant cell. As you study the figure, reflect on how each process fits into the big picture: As the most basic unit of living organisms, a cell performs all functions characteristic of life.

CONCEPT CHECK 10.6

- 1. MAKE CONNECTIONS** How do plants use the sugar they produce during photosynthesis to directly power the work of the cell? Provide some examples of cellular work. (See Figures 8.10, 8.11, and 9.5.)

For suggested answers, see Appendix A.

▼ **Figure 10.21 A review of photosynthesis.** This diagram shows the main reactants and products of photosynthesis as they move through the tissues of a tree (left) and a chloroplast (right).

➔ **Mastering Biology** HHMI Animation: Photosynthesis
MP3 Tutor: Photosynthesis

